

PHYS 496: Lecture # 1

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Lecture Span

1. Introduction to class
2. How to read physics papers
3. Writing effective titles

Introduction to class

1. Homework is due on gradescope
2. Email phys496@physics.illinois.edu if missing a class
3. Two colloquium reports due. One due near september. 2nd due in early november.

How to read physics papers

The 4 i's of paper reading.

1. Importance: does the paper contain relevant information for your research? Read title and abstract. Also, look at author list. The goal of every step should be to figure out if you're going to abandon this paper.
2. Iterate: Find main points of each section, also generate questions while reading. **REQUIRES ACTIVE BRAIN HOURS**. Also, figuring out how to understand a paper is good, like figuring out how the result of the paper changes if you change the underlying model, etc.
3. Interpret: write down key ideas in your own words.
4. Integrate: Figuring out how the information in the paper fits with your current knowledge.

Explaining something simple is the last step. Writing down things imprints it into the brain.

Writing effective titles

1. Make it interesting, but not too interesting. Keep them short.
2. Don't use qualitative words
3. Don't use names of people, places, etc.
4. Front load key words